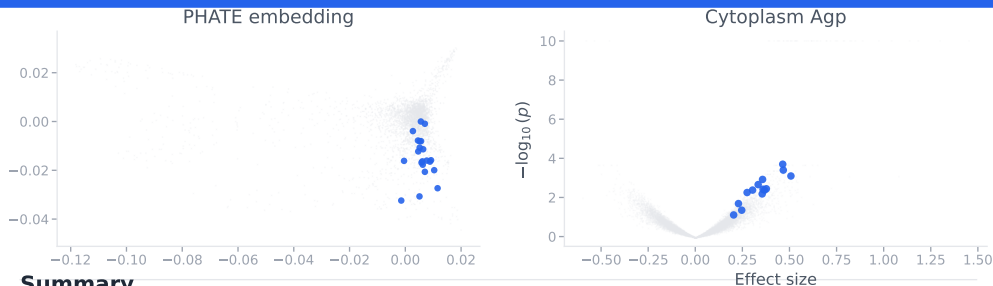


Cullin-RING E3 ubiquitin ligase (CRL) pathway

Cluster 10



This cluster shows HIGH confidence for the Cullin-RING E3 ubiquitin ligase (CRL) pathway, with ~75% of genes fitting this process. Core CRL machinery is present (CUL3, RBX1, UBE2M) along with multiple KCTD substrate adaptors (KCTD17, KCTD10, KCTD5). The morphological screen context (actin, membrane, organelle staining) aligns with known CRL functions in cytoskeletal regulation and protein turnover...

Established genes

CUL3, RBX1, UBE2M, KCTD17, KCTD10, KCTD5

Novel & uncharacterized genes

G6PC2 novel

Glucose-6-phosphatase involved in glucose metabolism. No clear connection to CRL pathway. Likely represents off-target or indirect morphological effect rather than novel pathway role.

TAOK3 novel

MAP kinase kinase kinase. While kinases can be CRL substrates, TAOK3 has established regulatory mechanisms. Clustering likely reflects downstream effects rather than novel direct regulation.

BTF3 novel

Transcription factor and nascent polypeptide-associated complex component. CRLs regulate transcription factors, but BTF3 connection is speculative and may reflect indirect effects.

CAB39 novel

STK11/LKB1 activator. Some kinase regulators are CRL substrates, but CAB39 regulation by ubiquitination is not well-documented. Connection may be indirect.

NCBP1 novel

Cap-binding complex component involved in mRNA processing. While CRLs regulate transcription/RNA processing factors, NCBP1 as a CRL substrate is not established but plausible given nuclear

NCBP2 novel

Cap-binding complex partner of NCBP1. Same rationale - potential novel CRL-mediated regulation of mRNA cap-binding.

CAPZA2 novel

Actin-capping protein. CRLs regulate cytoskeletal proteins, and morphological clustering suggests potential ubiquitin-mediated regulation of actin dynamics, though not previously documented for

CAPZB novel

Actin-capping protein partner of CAPZA2. Same rationale as CAPZA2 - potential novel CRL-mediated regulation of actin capping activity.

CFL1 novel

Cofilin, actin-depolymerizing factor. Known to be regulated by phosphorylation, but potential ubiquitin-mediated regulation by CRLs would be novel. Morphological phenotype overlap supports

+ 6 more